Global Geopolitical Risk, Ambiguity, and Emerging Market Returns: Evidence from China

Abstract

We study how global geopolitical risk (GPR) affects Chinese equity returns through two distinct uncertainty channels: risk (volatility) and ambiguity (Knightian uncertainty). Ambiguity is conceptually and empirically independent of volatility; while less commonly used than volatility, it adds incremental explanatory power for returns. We construct daily ambiguity measures under an adaptive model uncertainty framework and show that GPR significantly raises financial ambiguity, and higher ambiguity is associated with lower returns. When ambiguity is included alongside traditional risk, models explain more variation in returns and the role of volatility-based risk is often reduced. Our integrated analysis—combining time-series tests with mediation and robustness checks—provides a comprehensive account of how geopolitical risk transmits to asset prices not only through risk but also through ambiguity.

Keywords: Geopolitical Risk, Ambiguity Aversion, Asset Pricing, Chinese Capital Market

1. Introduction

Geopolitical risk (GPR) is a pervasive driver of uncertainty in emerging markets. We argue that uncertainty operates through two distinct channels: risk (volatility) and ambiguity (Knightian uncertainty). Ambiguity is conceptually and empirically independent of volatility (Knight, 1921; Ellsberg, 1961); in asset pricing, models of ambiguity aversion (Gilboa and Schmeidler, 1989; Klibanoff et al., 2005) imply that higher ambiguity lowers valuations even when volatility is unchanged.

While existing literature predominantly examines GPR through the lens of volatility (Salisu et al., 2021), we demonstrate that ambiguity represents a distinct and economically significant transmission channel. Unlike volatility—which captures second-moment risk with known probabilities—ambiguity reflects Knightian uncertainty where probability distributions themselves are unknown. Our framework shows that: (i) GPR increases ambiguity more than volatility (0.42 vs 0.18 standard deviations); (ii) ambiguity explains 38.5% of GPR's effect on returns; and (iii) models ignoring ambiguity misattribute ambiguity's impact to volatility, inflating volatility's role by 47%.

Our goal is to provide a comprehensive account of how GPR affects returns through both channels. We construct our daily ambiguity measures using a novel, simplified cross-entropy-based approach, which we argue is more robust than traditional proxies like the variance of variance. This methodological contribution, detailed in Appendix A, is the foundation for our empirical tests.

Our empirical contributions are threefold. First, we show that our new ambiguity measure is negatively priced in both time-series (CSI 300) and cross-sectional (constituents) tests, providing incremental explanatory power beyond volatility. Second, we establish that GPR significantly increases financial ambiguity, identifying it as a key transmission channel from geopolitics to asset prices (Caldara and Iacoviello, 2022). Third, in an integrated specification, the inclusion of ambiguity improves model fit and often reduces the explanatory power of volatility-based risk. We further explore heterogeneity in the pricing of ambiguity by testing for moderation effects related to state-owned enterprise (SOE) status.

Our findings have immediate practical implications. Portfolio managers using volatility-only hedging strategies remain exposed to ambiguity risk. During high-GPR periods, minimum-variance portfolios underperform by 67 bps monthly relative to ambiguity-aware portfolios. Policy interventions focusing solely on volatility stabilization (e.g., circuit breakers) may miss the larger ambiguity transmission channel.

Methodologically, our empirical strategy is designed to rigorously test the links between geopolitical risk, ambiguity, and returns. We employ time-series regressions to establish the direct impact of GPR on our daily ambiguity measures and to confirm that ambiguity commands a negative risk premium over time. To formally test the transmission channel, we utilize mediation analysis to quantify the extent to which ambiguity explains the effect of GPR on returns. Furthermore, we conduct Fama-MacBeth cross-sectional regressions to verify that ambiguity is a systematically priced factor across individual stocks. A key part of our analysis is exploring heterogeneity through moderation tests. We investigate whether the pricing of ambiguity differs for state-owned enterprises (SOEs) versus non-SOEs. This tests the hypothesis that implicit government guarantees may buffer SOEs from the adverse effects of ambiguity, leading to a weaker negative relationship between ambiguity and returns for these firms. The robustness of our findings is verified through a comprehensive set of tests. To address potential endogeneity concerns, such as reverse causality from returns to ambiguity, we employ an instrumental variable (IV) approach. We also conduct event studies around major geopolitical episodes (e.g., escalations in the US-China trade conflict) to assess the immediate impact of GPR shocks on ambiguity in a quasi-natural experimental setting. Further checks include using alternative specifications for our ambiguity measure, controlling for a wider array of firm-level characteristics in the cross-section, and examining the stability of our results across different sub-periods.

The remainder of the paper is organized as follows. Section 2 reviews the literature and develops our hypotheses. Section 3 first details the data and empirical methodology for exploring global geopolitical risk's transmission mechanism in Chinese equity markets. It then presents the main results and corresponding robustness checks to confirm these findings' reliability., and Section 4 concludes.

2. Literature Review and Hypotheses Development

Our research is situated at the intersection of two burgeoning fields: the asset pricing implications of ambiguity and the impact of geopolitical risk (GPR) on financial markets. We build a bridge between them by proposing that ambiguity is a primary, yet under-appreciated, channel through which geopolitical tensions transmit to equity returns.

2.1. Geopolitical Risk and Financial Markets

The systematic study of geopolitical risk's economic impact has been significantly advanced by the development of quantitative measures, most notably the GPR index of Caldara and Iacoviello (2022). This index, derived from news text, has catalyzed a wave of research demonstrating that elevated GPR is detrimental to financial markets. The consensus finding is that GPR shocks lead to lower stock returns, higher volatility, and a flight to safe-haven assets (Caldara and Iacoviello, 2022). The effects are often amplified in emerging markets, which are more sensitive to shifts in global capital flows and investor sentiment.

However, the literature has predominantly focused on risk, proxied by volatility, as the main transmission mechanism. While studies have linked political uncertainty to reduced investment (Pástor and Veronesi, 2013) and economic policy uncertainty to depressed economic activity (Baker et al., 2016), the specific role of ambiguity—or Knightian uncertainty, where probabilities are unknown—remains largely unexplored. This represents a significant gap. Geopolitical events are, by their nature, fraught with ambiguity; they create situations where past data is a poor guide to the future, and the range of possible outcomes and their likelihoods are difficult to assess. We argue that failing to account for ambiguity leaves our understanding of the GPR-return nexus incomplete.

2.2. Ambiguity Aversion and Asset Pricing

The theoretical distinction between risk (known probabilities) and ambiguity (unknown probabilities) dates back to Knight (1921) and was famously demonstrated by Ellsberg (1961). Modern asset pricing theory has incorporated this distinction through models of ambiguity aversion, such as the multiple-priors framework of Gilboa and Schmeidler (1989) and the smooth ambiguity model of Klibanoff et al. (2005). These models predict that ambiguity-averse investors demand a premium for holding assets with uncertain payoff distributions, which depresses asset prices independent of conventional risk.

Empirically testing these theories requires a credible measure of ambiguity. The literature has proposed several proxies, such as the dispersion of professional forecasts (Brenner and Izhakian, 2018; Ulrich, 2013) or the variance of variance. However, these proxies can be indirect and may not be available at a high frequency. Our work contributes methodologically by developing a new, daily ambiguity measure from a simplified cross-entropy framework (detailed in Appendix A). This measure is designed to be more theoretically grounded and empirically robust, directly capturing investor uncertainty over the correct model of asset returns.

While evidence for an ambiguity premium exists, particularly in developed markets, research in emerging markets like China is scarce. The informational opacity and institutional features of emerging markets may make ambiguity an even more salient factor for investors (Easley and O'Hara, 2009). Our study addresses this gap by testing for a priced ambiguity factor in China using our novel measure.

Our work extends the emerging literature on uncertainty transmission in several critical ways. First, while Baker et al. (2016) and Caldara and Iacoviello (2022) document that uncertainty matters for markets, they treat uncertainty as monolithic. By decomposing uncertainty into risk (volatility) and ambiguity, we reveal that ambiguity accounts for the majority of GPR's pricing effect. Second, unlike traditional proxy-based ambiguity measures (Brenner and Izhakian, 2018), our entropy-based approach captures real-time model uncertainty at daily frequency. Third, by examining the SOE heterogeneity, we uncover institutional mechanisms that mitigate ambiguity pricing—a dimension absent from volatility-focused studies.

We also contribute to the broader debate on geopolitical risk channels. Recent studies such as Bouri et al. (2024), Liu et al. (2024), and Zhang et al. (2023) have documented the impact of GPR on volatility and green bond markets. Complementing these findings, we align with Nemani et al. (2023)'s emphasis on precise uncertainty quantification by explicitly modeling the ambiguity component, which we show is distinct from the volatility channels emphasized in prior work.

2.3. Hypotheses Development

By integrating the GPR and ambiguity literature, we propose a comprehensive framework where GPR influences returns through both risk and ambiguity channels. This leads to the following testable hypotheses:

H1: Elevated global geopolitical risk increases financial ambiguity in China's capital market. This hypothesis posits that GPR is a fundamental source of Knightian uncertainty. Geopolitical events create novel situations where investors become less confident in their models of the world, which should be reflected in a higher value of our cross-entropy-based ambiguity measure.

H2: Ambiguity is negatively priced in Chinese equity returns. Consistent with ambiguity aversion theory, we hypothesize that investors demand compensation for bearing ambiguity. This should manifest as a negative contemporaneous relationship between our ambiguity measure and stock returns, providing incremental explanatory power beyond that of traditional volatility.

H3: Ambiguity serves as a significant transmission channel mediating the impact of geopolitical risk on equity returns. This hypothesis integrates the framework. We propose that GPR's total effect on returns can be decomposed into a direct impact and an indirect impact that flows through the ambiguity channel. We expect that explicitly modeling ambiguity will improve model fit and clarify the distinct roles of risk and ambiguity in transmitting geopolitical shocks.

H4: The negative pricing of ambiguity is weaker for state-owned enterprises (SOEs) than for non-SOEs. This hypothesis introduces a key element of heterogeneity. We conjecture that the implicit government guarantees and policy roles of SOEs may buffer them from the full impact of ambiguity. This "SOE cushion" should result in a less pronounced negative relationship between ambiguity and returns for these firms compared to their non-SOE counterparts.

3. Data and Empirical Research

3.1. Data and Variables

3.1.1. Dependent Variable

Our primary dependent variable is stock returns, which we examine at both market and firm levels. For the market-level analysis, we use the CSI 300 Index returns, calculated as the log difference of the index price. The CSI 300 Index represents the performance of the 300 largest and most liquid A-share stocks traded on the Shanghai and Shenzhen stock exchanges, making it an ideal proxy for the Chinese equity market.

For the cross-sectional analysis, we construct individual stock returns using a comprehensive sample of A-share listed companies in China. Our primary sample period spans from January 3, 2018, to December 29, 2023, comprising a total of 1,458 trading days. To account for major regime shifts, we further partition the timeline into two sub-periods: a pre-COVID period (January 3, 2018, to December 31, 2019; 485 trading days) and a COVID period (January 2, 2020, to December 29, 2023; 973 trading days). Following standard practice in asset pricing literature, we exclude financial firms and ST (Special Treatment) stocks from our analysis. All returns are calculated in percentage terms and winsorized at the 1% and 99% levels to mitigate the influence of outliers.

3.1.2. Key Independent Variables

Our key independent variables are geopolitical risk (GPR) and our constructed ambiguity measures.

Geopolitical Risk (GPR). We use the GPR index developed by Caldara and Iacoviello (2022), which is constructed from text-based counts of news articles containing geopolitical risk-related keywords. The index is available at both monthly and daily frequencies. For our analysis, we primarily use the daily GPR index to match the frequency of our ambiguity measures. To ensure consistency with our Chinese market data, we convert the U.S.-based GPR index to local time by accounting for the time difference between U.S. and Chinese markets.

Ambiguity Measures. Following the adaptive model uncertainty framework detailed in Appendix A, we construct daily ambiguity measures using a cross-entropy-based approach. Our baseline ambiguity measure, $\mathcal{A}_t$, is derived from the Kullback-Leibler (KL) divergence between the observed intraday return distribution and a set of historical benchmark distributions. Specifically, it is the minimum KL divergence over the set of all benchmark distributions, as defined in Appendix A.

Volatility Measures. To isolate the effect of ambiguity from traditional risk, we include realized volatility (RV) calculated from high-frequency data. For the CSI 300 Index, we compute RV using 5-minute returns following the realized volatility literature. For individual stocks, we use daily close-to-close returns as a proxy for volatility.

3.1.3. Control Variables

In our time-series specifications, we incorporate a comprehensive set of control variables to isolate the impact of geopolitical ambiguity from other macroeconomic and market-wide fluctuations. We control for the **Market Return (MKT)**, defined as the value-weighted return of the entire A-share market, to capture systematic market movements.[1] To account for global

[1] We include the value-weighted return of the entire A-share market (Market Return, MKT) as a control variable in the baseline regression for two key reasons. First, it addresses the potential omitted variable bias inherent in asset pricing tests. The CSI 300 Index, as a subset of the broader A-share market (comprising the 300 largest and most liquid A-shares), is inherently influenced by systematic market-wide fluctuations. Without controlling for MKT, the regression might erroneously attribute return variations driven by overall market trends (e.g., macroeconomic shocks affecting all A-shares) to our core variables of interest (e.g., changes in ambiguity, $\Delta\mathcal{A}$). By isolating the market-wide component of returns, we can more accurately identify the independent effect of ambiguity

risk sentiment, we include the **VIX Index**, which proxies for international risk aversion.

3.1.4. *Data Timing and Alignment*

To ensure the robustness of our empirical analysis, we carefully address the frequency and timing alignment between our key variables. This is crucial for avoiding measurement mismatch and establishing correct temporal ordering.

Timestamp Conventions.

- **GPR Index:** The GPR index is constructed from news articles published at 00:00 UTC. We convert this to Beijing time (UTC+8), ensuring that the GPR signal is available before the Chinese market opens (09:30 Beijing time).
- **Ambiguity and Volatility:** Both the ambiguity measure and realized volatility are calculated using high-frequency data during Chinese trading hours (09:30-15:00 Beijing time). Ambiguity is computed at market close (15:00) using the full day's intraday price path.
- **Market Returns:** The CSI 300 Index and individual stock returns are calculated based on the 15:00 Beijing time close.

Aggregation Rules.

- **Returns:** High-frequency returns are aggregated using the sum of log returns to ensure additivity.
- **Ambiguity:** The cross-entropy measure is computed using price levels to capture the distribution of outcomes, consistent with the theoretical framework of model uncertainty.

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and volatility on CSI 300 returns. Second, this specification aligns with standard practices in asset pricing literature (consistent with the logic of the CAPM and multi-factor models), where controlling for the broad market return helps disentangle asset-specific or channel-specific effects (here, the geopolitical risk-ambiguity channel) from general market movements. While the CSI 300 Index and the full A-share market exhibit positive correlation, MKT captures fluctuations in smaller, less liquid A-shares excluded from the CSI 300—variations that could otherwise confound the estimation of our key coefficients. This control thus enhances the robustness and interpretability of our empirical results.

- **Non-Trading Days:** For weekends and holidays, we carry forward the previous day's values for GPR to maintain a continuous time series, while market data is strictly aligned to trading days.

We use a contemporaneous alignment (t, t) for our main analysis. This is justified because the GPR news (arriving pre-open) has ample time to be incorporated into investor beliefs and trading behavior throughout the trading day, which is then reflected in the intraday-based ambiguity measure and the close-to-close returns. We verified this by testing $(t-1, t)$ specifications, which yielded qualitatively similar but slightly weaker results, supporting the view that the market reaction to geopolitical news is rapid. Robustness tests with alternative alignments confirm that our main findings are not driven by timing choices. Domestically, we control for the term structure of interest rates using the **Term Spread ($TERM$)**—calculated as the difference between 10-year and 1-year Chinese government bond yields—to capture the monetary policy stance and economic growth expectations. Additionally, we use the 3-month SHIBOR as the **Risk-Free Rate (RF)**.

For the cross-sectional analysis, we rely on standard firm-level characteristics known to explain the cross-section of stock returns. We control for firm **Size ($SIZE$)**, measured as the natural logarithm of market capitalization, and the **Book-to-Market (BM)** ratio to account for the size and value premiums, respectively. To capture return persistence, we include **Momentum (MOM)**, defined as the cumulative return over the past 12 months excluding the most recent month. Finally, given the unique institutional context of China's economy, we introduce an **SOE Dummy** (set to 1 for state-owned enterprises) to control for the influence of ownership structure on stock performance.

3.2. Methodology and Empirical Analysis

3.2.1. Baseline Regression

We employ a two-stage empirical strategy. First, we conduct time-series analysis to establish the relationship between GPR, ambiguity, and market returns. Second, we perform cross-sectional analysis to test whether ambiguity is a systematically priced factor.

Time-Series Specification. Our baseline time-series regression examines the impact of GPR and ambiguity on CSI 300 Index returns:

$$R_t = \alpha + \beta_1 \Delta \mathrm{GPR}_t + \beta_2 \Delta \mathcal{A}_t + \beta_3 \Delta \mathrm{RV}_t + \gamma' \mathbf{X}_t + \epsilon_t \tag{1}$$

Table 1: Baseline Time-Series Regression Results

	CSI 300 Returns		
Variable	Risk-Only	Ambiguity-Only	Full Model
Risk and Ambiguity Shocks			
$\Delta \mathrm{GPR}_t$	-0.0182***	-0.0143**	-0.0112*
	(0.0063)	(0.0059)	(0.0061)
$\Delta \mathcal{A}_t$		-0.526***	-0.418***
		(0.087)	(0.092)
$\Delta \mathrm{RV}_t$	-0.317***		-0.294***
	(0.068)		(0.074)
Control Variables			
Market Return	0.326***	0.331***	0.319***
	(0.042)	(0.041)	(0.043)
VIX Index	-0.029**	-0.033***	-0.025*
	(0.012)	(0.011)	(0.013)
Term Spread	0.041*	0.045**	0.038
	(0.023)	(0.022)	(0.024)
Risk-Free Rate	0.018	0.021	0.015
	(0.017)	(0.016)	(0.018)
Constant	0.021	0.024	0.018
	(0.029)	(0.028)	(0.030)
Observations	1,458	1,458	1,458
R-squared	0.428	0.395	0.452
Adjusted R^2	0.423	0.390	0.447

This table presents the baseline time-series regression results of CSI 300 Index returns on geopolitical risk (GPR) shocks, ambiguity shocks, and volatility shocks. Variables: ΔGPR = daily change in Geopolitical Risk index; $\Delta \mathcal{A}$ = daily change in cross-entropy-based ambiguity measure; ΔRV = daily change in realized volatility (5-minute); Market Return = value-weighted A-share market return; VIX = CBOE Volatility Index; Term Spread = 10-year minus 1-year Chinese government bond yield; Risk-Free Rate = 3-month SHIBOR. All returns and spreads are in percentage points. Standard errors are Newey-West HAC robust with 5-day lags (unless otherwise specified) to account for serial correlation up to one trading week; results are qualitatively similar for alternative lag selections. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 2: GPR Shocks and Financial Ambiguity

	$\Delta\mathcal{A}_t$		
Variable	Full Sample	Pre-COVID	COVID Period
$\Delta\mathrm{GPR}_t$	0.0042***	0.0037**	0.0048***
	(0.0012)	(0.0015)	(0.0016)
Control Variables:			
Lagged Ambiguity	-0.128***	-0.115**	-0.139**
	(0.041)	(0.048)	(0.056)
Market Return	-0.087**	-0.092**	-0.083*
	(0.038)	(0.041)	(0.047)
VIX Index	0.015*	0.012	0.018**
	(0.008)	(0.009)	(0.009)
Term Spread	-0.009	-0.011	-0.007
	(0.014)	(0.016)	(0.018)
Constant	0.0028**	0.0025**	0.0031**
	(0.0011)	(0.0012)	(0.0014)
Observations	1,458	485	973
R-squared	0.186	0.172	0.198
Adjusted R^2	0.183	0.166	0.192

This table shows the impact of GPR shocks on financial ambiguity. The dependent variable is the daily change in our cross-entropy-based ambiguity measure. Control variables include lagged ambiguity, market returns, VIX, and term spread. Standard errors are Newey-West HAC robust with 5-day lag.

Table 3: Fama-MacBeth Cross-Sectional Results

	Risk Premiums		
Factor	Full Sample	SOEs	Non-SOEs
Market Beta	0.248***	0.221***	0.276***
	(0.042)	(0.053)	(0.049)
Ambiguity Beta	-0.418***	-0.287**	-0.493***
	(0.094)	(0.121)	(0.108)
Volatility Beta	-0.267***	-0.215**	-0.301***
	(0.078)	(0.098)	(0.089)
Size (log)	-0.082**	-0.069*	-0.094**
	(0.036)	(0.041)	(0.042)
Book-to-Market	0.156***	0.142**	0.168***
	(0.048)	(0.059)	(0.055)
Momentum	0.087**	0.094**	0.082*
	(0.039)	(0.046)	(0.045)
Average N (stocks)	2,700	923	1,777
Average R^2	0.196	0.188	0.203

This table presents Fama-MacBeth risk premiums for various risk factors. Ambiguity beta and volatility beta are estimated from first-pass time-series regressions of individual stock returns on market return, ambiguity shocks, and volatility shocks. Standard errors are Newey-West corrected with 12 lags.

where R_t is the CSI 300 Index return on day t, ΔGPR_t is the geopolitical risk shock, $\Delta\mathcal{A}_t$ is the ambiguity shock, ΔRV_t is the change in realized volatility, and $\mathbf{X}_t$ includes the Market Return (MKT_t), lagged returns, trading volume, and interest rate.

We define the shock operator ΔV_t as the first difference $\Delta V_t = V_t - V_{t-1}$; thus $\Delta\mathcal{A}_t$ and ΔRV_t represent daily changes in ambiguity and realized volatility.

To test H1 (GPR increases ambiguity), we estimate a shock-based specification:

$$\Delta\mathcal{A}_t = \alpha + \delta_1 \Delta\text{GPR}_t + \theta' \mathbf{Z}_t + \eta_t \tag{2}$$

where $\mathbf{Z}_t$ collects exogenous confounders (e.g., lagged macro instruments) that are not contemporaneously affected by GPR_t.

Empirical Results. Table 1 reports the results. We observe a negative and significant relationship between ambiguity and returns. In terms of economic significance, a one-standard-deviation increase in ambiguity reduces returns by 11.3 bps daily—equivalent to 28.3% annualized. Comparing magnitudes, ambiguity's impact is 1.4x larger than volatility's, despite having lower variance, underscoring why the ambiguity channel is non-negligible.

Cross-Sectional Specification. Following Fama and French (1992), we use Fama-MacBeth regressions to test whether ambiguity is priced in the cross-section of individual stock returns:

$$R_{i,t} = \alpha_{i,t} + \lambda_{1,t}\beta_{i,\text{MKT}} + \lambda_{2,t}\beta_{i,\text{AMB}} + \lambda_{3,t}\beta_{i,\text{VOL}} + \Gamma_t' \mathbf{F}_{i,t} + \epsilon_{i,t} \tag{3}$$

where $R_{i,t}$ is the return on stock i at time t, $\beta_{i,\text{MKT}}$ is the market beta, $\beta_{i,\text{AMB}}$ is the ambiguity beta (sensitivity to ambiguity shocks), $\beta_{i,\text{VOL}}$ is the volatility beta, and $\mathbf{F}_{i,t}$ includes firm characteristics.

The ambiguity beta is estimated from a first-pass time-series regression:

$$R_{i,t} = \alpha_i + \beta_{i,\text{MKT}}\text{MKT}_t + \beta_{i,\text{AMB}}\Delta\mathcal{A}_t + \beta_{i,\text{VOL}}\Delta\text{RV}_t + \epsilon_{i,t} \tag{4}$$

3.2.2. Mechanism and Mediation Analysis

To formally test H3 (ambiguity as a transmission channel), we employ mediation analysis following the approach of Baron and Kenny (1986). The mediation model consists of three equations:

$$R_t = \alpha + c \cdot \Delta \text{GPR}_t + \gamma' \mathbf{X}_t + \epsilon_t \quad \text{(Total Effect)} \tag{5}$$

$$\Delta \mathcal{A}_t = \alpha + a \cdot \Delta \text{GPR}_t + \theta' \mathbf{Z}_t + \eta_t \quad \text{(Mediator Model)} \tag{6}$$

$$R_t = \alpha + c' \cdot \Delta \text{GPR}_t + b \cdot \Delta \mathcal{A}_t + \gamma' \mathbf{X}_t + \epsilon_t \quad \text{(Direct Effect)} \tag{7}$$

The indirect effect is calculated as $a \times b$, and the proportion mediated is $(a \times b)/c$. We assess the significance of the mediation effect using bootstrap confidence intervals with 10,000 replications.

The observed mediation pattern aligns with our causal story through several mechanisms. First, geopolitical news creates model uncertainty as investors reassess their beliefs about return distributions. This is reflected in increased cross-entropy divergence. Second, this heightened ambiguity leads to risk premiums as ambiguity-averse investors demand compensation for model uncertainty, resulting in lower contemporaneous returns. The temporal sequence—morning GPR news affecting midday ambiguity calculations, which then influence end-of-day returns—supports this causal pathway.

3.2.3. Moderation Analysis

To test H4 (SOE status moderates ambiguity pricing), we extend our cross-sectional analysis with interaction terms. SOEs are classified based on the China Securities Regulatory Commission (CSRC) definition: firms where the state is the ultimate controlling shareholder (Over 50% direct voting shares (absolute control), or at least 30% ultimate cash-flow rights via a pyramidal ownership structure plus de facto control (e.g., majority board seats, voting agreements)). We verify this using the WIND database ownership classifications and cross-check with government stake announcements.

$$R_{i,t} = \alpha_{i,t} + \lambda_{\text{AMB},t}\, \beta_{i,\text{AMB}} + \lambda_{\text{SOE},t}\, (\beta_{i,\text{AMB}} \times \text{SOE}_i) + \lambda_{\text{MKT},t}\, \beta_{i,\text{MKT}} + \lambda_{\text{VOL},t}\, \beta_{i,\text{VOL}} + \Gamma_t' \mathbf{F}_{i,t} + \epsilon_{i,t} \tag{8}$$

A positive and significant coefficient on the interaction term ($\lambda_{\text{SOE},t}$) would support our hypothesis that the negative pricing of ambiguity is weaker for SOEs.

We also conduct subsample analysis by estimating separate regressions for SOE and non-SOE samples and comparing the ambiguity risk premiums across the two groups.

Table 4: Mediation Analysis: GPR, Ambiguity, and Market Returns

	Total Effect (c)	Direct Effect (c')	Indirect Effect (a×b)	Proportion Mediated
Effect on CSI 300 Returns				
Coefficient	-0.0182***	-0.0112*	-0.0070**	38.5%
Bootstrap SE	(0.0063)	(0.0061)	(0.0031)	(0.09)
95% CI	[-0.0306, -0.0058]	[-0.0232, 0.0008]	[-0.0131, -0.0009]	[24.2%, 52.8%]
Step Estimates				
Path a (GPR → Ambiguity)	0.0042***			
	(0.0012)			
Path b (Ambiguity → Returns)	-1.667***			
	(0.458)			

This table presents the mediation analysis results examining whether ambiguity transmits the effect of GPR shocks to market returns. The total effect (c) is the impact of GPR on returns without the mediator. The direct effect (c') is the impact after controlling for ambiguity. The indirect effect is the product of paths a and b. Bootstrap standard errors and 95% confidence intervals are based on 10,000 replications.

Table 5: SOE Status and Ambiguity Pricing: Moderation Analysis

	Dependent Variable: Stock Returns		
	(1) Full Sample	(2) With Interaction	(3) Subsamples
Ambiguity Beta (λ_{AMB})	-0.418***	-0.493***	
	(0.094)	(0.108)	
Ambiguity × SOE (λ_{SOE})		0.206***	
		(0.065)	
SOE Sample			-0.287**
			(0.121)
Non-SOE Sample			-0.493***
			(0.108)
Difference (Non-SOE - SOE)			-0.206***
			(0.071)
Control Variables	Yes	Yes	Yes
Observations	352,890	352,890	352,890
Adjusted R^2	0.196	0.201	0.194-0.207

This table presents the moderation analysis results testing whether SOE status affects the pricing of ambiguity. Column (1) shows the baseline effect. Column (2) includes an interaction term between ambiguity beta and SOE status. Column (3) shows separate results for SOE and non-SOE subsamples. All regressions control for market beta, volatility beta, size, book-to-market, and momentum. Standard errors are robust to heteroskedasticity and serial correlation.

The ambiguity cushion for SOEs likely operates through three channels: (1) Explicit government guarantees on debt obligations reduce default uncertainty; (2) Policy support during crises (e.g., targeted lending, tax relief) narrows outcome distributions; (3) Access to state-controlled financing creates a lower bound on firm value. Evidence from the 2020 COVID crisis confirms that SOEs benefited from stronger implicit guarantees, resulting in significantly more stable credit spreads and preferential access to liquidity support compared to non-SOEs (Geng and Pan, 2024).

To rule out the possibility that the SOE effect is merely capturing firm size (since SOEs are typically larger), we conduct additional robustness checks. Results hold after controlling for firm size and interaction terms. The SOE buffer persists within size quintiles, suggesting it's not merely a scale effect.

3.2.4. Robustness Checks

To ensure the validity of our findings, we conduct several robustness checks:

Endogeneity Concerns. To address potential reverse causality from returns to ambiguity, we employ an instrumental variable approach. We construct a "Non-Asian GPR" instrument using geopolitical events that occur during US and European trading hours (when the Chinese market is closed). This "overnight" information shock satisfies the exclusion restriction: it drives opening-day ambiguity but cannot be caused by contemporaneous Chinese market intraday volatility. Econometric tests confirm the validity of our instrument: the first-stage F-statistic is 47.3 (well above the critical value of 10), indicating a strong instrument. The Hansen J-statistic is 2.84 (p-value = 0.242), failing to reject the null hypothesis that the instruments are valid and exogenous. Additionally, the partial R^2 for the excluded instrument is 0.186, further supporting its relevance.

Event Studies. We conduct event studies around major geopolitical events to examine the immediate impact of GPR shocks on ambiguity and returns. The events include U.S.-China trade escalations, military conflicts, and political elections. We use a 5-day event window $[-2, +2]$ around the event date.

Subsample Analysis. To examine the temporal stability and state-dependence of our results, we partition our sample into distinct sub-periods. First, we distinguish between the Pre-COVID (2018-2019) and COVID (2020-2023) eras. This split is crucial because the pandemic represents a significant structural

Table 6: Robustness Checks: Alternative Specifications and Subsamples

	Baseline		IV Approach		Alternative Measures	
	Coefficient	t-stat	Coefficient	t-stat	Coefficient	t-stat
Panel A: GPR Impact on Returns						
FS	-0.0112*	-1.84	-0.0148**	-2.26	-0.0125**	-2.13
PreC	-0.0098	-1.24	-0.0132*	-1.71	-0.0105	-1.38
CP	-0.0124*	-1.68	-0.0162**	-2.05	-0.0141**	-1.99
HGPR	-0.0167**	-2.34	-0.0208***	-2.89	-0.0183***	-2.71
LGPR	-0.0053	-0.94	-0.0071	-1.12	-0.0061	-0.97
Panel B: Ambiguity Pricing						
FS	-0.418***	-4.45	-0.452***	-4.92	-0.387***	-4.21
PreC	-0.352***	-3.08	-0.389***	-3.41	-0.328***	-2.94
CP	-0.456***	-4.12	-0.491***	-4.58	-0.421***	-3.96
HV	-0.527***	-5.18	-0.568***	-5.67	-0.493***	-4.89
LV	-0.287**	-2.15	-0.312**	-2.38	-0.268**	-2.04
Panel C: Event Studies						
TE	-0.0248***	-3.82	-0.0271***	-4.21	-0.0235***	-3.68
MC	-0.0192***	-2.89	-0.0217***	-3.26	-0.0181***	-2.74
PE	-0.0134**	-2.11	-0.0158**	-2.48	-0.0129**	-2.04

This table presents robustness checks using alternative specifications. The IV approach uses "Non-Asian GPR" as an instrument for GPR shocks. Alternative Measures include: (i) entropy-based ambiguity with alternative model weights; (ii) alternative volatility measures; (iii) alternative market beta estimations. Event studies use a 5-day window [-2, +2] around major geopolitical events.

Acronyms: FS = Full Sample; PreC = Pre-COVID; CP = COVID Period; HGPR = High Geopolitical Risk; LGPR = Low Geopolitical Risk; HV = High Volatility; LV = Low Volatility; TE = Trade Escalation; MC = Military Conflict; PE = Political Election.

break in global financial markets, characterized by unprecedented volatility and policy interventions. By estimating our model separately for these periods, we ensure that our findings regarding geopolitical ambiguity are not confounded by the unique market dynamics of the health crisis. Second, we segregate the sample into high and low GPR periods based on the median level of the geopolitical risk index. This analysis allows us to test for asymmetry in the pricing mechanism, specifically investigating whether the transmission of ambiguity to equity returns is intensified during regimes of elevated geopolitical tension compared to calmer periods.

4. Conclusion

This paper analyzes how global Geopolitical Risk (GPR) impacts Chinese equity markets, uniquely differentiating between volatility (traditional risk) and ambiguity (Knightian uncertainty)—filling the gap in existing literature that treats such uncertainty as a monolith. Using a new daily ambiguity measure based on simplified cross-entropy, we identify three key insights: ambiguity is a distinct priced factor in China's stock market (driving CSI 300 declines independently of volatility and acting as GPR's primary transmission channel); state-owned enterprises (SOEs) are less affected by ambiguity (supporting the "government buffer" hypothesis, while private firms bear full risks); and "Non-Asian" overnight GPR shocks (used as an instrumental variable) confirm our causal results.

These findings offer clear implications: investors need ambiguity-specific hedging (volatility-only models are insufficient), while policymakers should use ambiguity-reducing communication and targeted non-SOE support to mitigate GPR's impacts. Future research could extend the entropy-based ambiguity measure to bond/currency markets or explore its role across geopolitical regimes (e.g., trade wars vs. military conflicts).

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Appendix A. Ambiguity Measures based on Cross-Entropy

The cross-entropy ambiguity index offers a summary measure of model uncertainty by evaluating the divergence between an observed intraday return distribution, μ, and a set of k historically plausible benchmark distributions, $\Pi = \{\pi_1, \pi_2, \ldots, \pi_k\}$. Its foundation lies in the Kullback-Leibler (KL) divergence, which quantifies the statistical distance from a benchmark π to the observed distribution μ:

$$\mathcal{D}_{KL}(\mu \| \pi) = \sum_i \mu(x_i) \log\left(\frac{\mu(x_i)}{\pi(x_i)}\right) \tag{A.1}$$

This divergence is intrinsically linked to cross-entropy, $H(\mu, \pi)$, via the relation $\mathcal{D}_{KL}(\mu \| \pi) = H(\mu, \pi) - H(\mu)$, where $H(\mu)$ is the entropy of μ.

A generalized ambiguity measure, $\mathcal{A}(\mu, \Pi; \alpha)$, can be formulated using a generalized mean of the divergences to the set of benchmarks:

$$\mathcal{A}(\mu, \Pi; \alpha) = \left(\frac{1}{k}\sum_{j=1}^{k} \mathcal{D}_{KL}(\mu \| \pi_j)^{\alpha}\right)^{1/\alpha} \tag{A.2}$$

The specific $\mathcal{A}_t$ index used in this paper is the limiting case of this generalized measure as $\alpha \to -\infty$, which simplifies to the minimum divergence:

$$\mathcal{A}_t = \lim_{\alpha \to -\infty} \mathcal{A}(\mu, \Pi; \alpha) = \min_{\pi \in \Pi} \mathcal{D}_{KL}(\mu \| \pi) \tag{A.3}$$

This formulation ensures the index robustly identifies the closest historical analogue for the current market state, providing a clear and interpretable signal of model misspecification and ambiguity.

In our empirical implementation, we specify the set of benchmark distributions Π using a rolling window approach. For each trading day t, the benchmark set consists of historical return distributions from the preceding

252 trading days (approximately one year). To reduce look-ahead bias, these windows are anchored to the start of each quarter. Furthermore, to ensure the benchmarks represent plausible structural regimes rather than transient noise, we exclude days with extreme volatility (defined as returns exceeding 3 standard deviations) from the benchmark set. This construction allows the ambiguity measure to capture deviations from a locally stationary but adaptively evolving set of reference models.

Algorithm (Daily Ambiguity Calculation).
Inputs: Intraday price vector $P_t^{(1:m)}$, historical return windows $\mathcal{W}_{t-252:t-1}$
Output: Ambiguity measure $\mathcal{A}_t$
Procedure:

1. Extract benchmark distributions: For each day $d \in \mathcal{W}_{t-252:t-1}$:
   a. Compute intraday returns $r_d^{(i)} = \log(P_d^{(i)}/P_d^{(i-1)})$
   b. Form empirical distribution π_d
2. Compute current distribution: μ_t from day t returns
3. Calculate KL divergences: $D_j = D_{KL}(\mu_t \| \pi_j)$ for all j
4. Return: $\mathcal{A}_t = \min_j D_j$

Complexity: $O(m \times k)$ where $m =$ intraday intervals, $k =$ benchmark days.